

Welcome back, Caden

Technical Interview Prep · Week 2

C

Current streak

ACTIVE

6 days

Problems solved

42 / 200

Readiness score

61%

Interview loop

14 days left

Today's plan

3 items · ~1h 45m

Lesson · Graph traversal foundations

IN PROGRESS

01 Number of Islands

NOT STARTED

02 Course Schedule

NOT STARTED

Resume lesson →

Track progress

Technical Interview Prep 38 / 180

OA Prep 4 / 20

Priority patterns

1 Dynamic Programming

2 Graphs

3 Trees

4 Union-Find

Study plan

Data-driven curriculum · ordered by interview priority

- OA Prep
- Technical Interview Prep

01	Dynamic Programming 1D & 2D · 30 problems · extended lesson TOP PRIORITY Mediums 6/30 LEARNING
02	Graphs BFS / DFS / Dijkstra · 30 problems · extended lesson TOP PRIORITY Mediums 12/30 LEARNING
03	Trees Traversal, BFS/DFS · 20 problems Easy 0/20 NOT STARTED
04	Union-Find / DSU Highest complexity ceiling · 12 problems Easy 0/12 NOT STARTED
05	Arrays & Hashing Foundational · 20 problems Mediums 18/20 MASTERED

Concept

Worked Example

Common Mistakes

Complexity

Grid traversal with DFS

A graph doesn’t need an explicit adjacency list — a 2D grid is one implicitly, where each cell connects to its up/down/left/right neighbors. Number of Islands asks you to count connected components of land cells, which is the same traversal you’d run on any graph: visit a cell, mark it, and recurse into unvisited neighbors that satisfy the condition.

PATTERN SKELETON

```
def dfs(r, c):
    if out_of_bounds(r, c) or visited[r][c]:
        return
    visited[r][c] = True
    for dr, dc in DIRECTIONS:
        dfs(r + dr, c + dc)
```

Why this matters for the interview

This exact skeleton generalizes to ~90% of grid and adjacency-list graph questions you’ll see. Interviewers weight your ability to state the base case and visited-check out loud before coding — narrate it the same way every time.

```
1 class Solution:
2     def numIslands(self, grid: List[List[str]]) -> int:
3     # base cases + visited set
4         rows, cols = len(grid), len(grid[0])
5         visited = set()
6         islands = 0
7
8         def dfs(r, c):
9             if (r, c) in visited or ...
10            return
11            visited.add((r, c))
```

TEST RESULTS

✓ Case 1 · grid = `[["1","1","0"]]`4ms

☐ Case 2 · single-cell grid

☐ Case 3 · all water

Progress

Session history, consistency, and mock interview log

Problems solved · last 14 days

+42 total



Sep 3

Sep 10

Sep 16

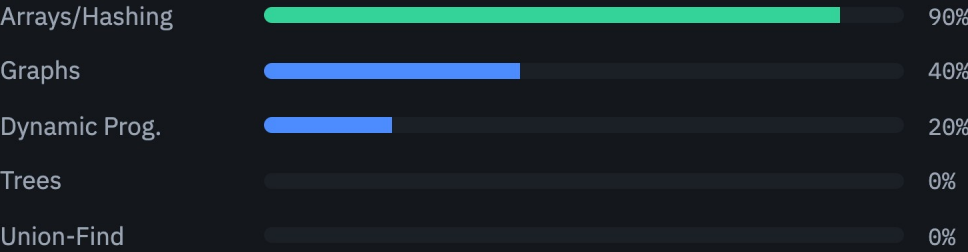
Consistency



6-day streak

Best: 9 days

Mastery by pattern



Mock interview log

DATE	PATTERN	RESULT
Sep 14	Graphs	SOLVED
Sep 11	Arrays	SOLVED
Sep 9	DP · 1D	TIMED OUT
Sep 6	Graphs	PARTIAL